Exp. 10 Creating CI/CD Pipeline in Azure DevOps

AIM

To create and configure a Continuous Integration and Continuous Deployment (CI/CD) pipeline in Azure DevOps for automatic build and deployment of an application.

TOOLS REQUIRED

- Microsoft Azure DevOps
- GitHub or Azure Repos
- Web Browser
- Azure Subscription

PREREQUISITES

- Project source code available in GitHub
- Azure Web App already created
- Azure DevOps account and project created

PROCEDURE

Step 1: Create Azure DevOps Project

- Login to Azure DevOps portal
- Click on **New Project**
- Enter:
    - Project Name (e.g., AttendanceAppPipeline)
    - Visibility: Private/Public
- Click **Create**

Step 2: Connect Repository

- Go to **Pipelines → Create Pipeline**
- Select source:
  - GitHub (or Azure Repos)
- Authorize access to GitHub
- Select your repository:
  - *Project*

Step 3: Configure Pipeline

- Choose pipeline configuration:
  - YAML file (recommended)
- Azure will auto-detect environment (Node.js / Python)

Example YAML:

```
trigger:
- main

pool:
  vmImage: 'ubuntu-latest'

steps:
- task: UseNode@1
  inputs:
    version: '18.x'

- script: |
    npm install
    npm run build
  displayName: 'Install and Build'

- task: AzureWebApp@1
  inputs:
    azureSubscription: 'Your-Azure-Connection'
    appName: 'quizapp-online'
    package: '.'
```

Step 4: Run Pipeline

- Click **Run Pipeline**
- Pipeline will:
  - Fetch code
  - Install dependencies
  - Build application
  - Deploy to Azure

Step 5: Monitor Pipeline

- View pipeline execution in **Pipelines → Runs**
- Check:
  - Build logs
  - Deployment status
- Fix errors if any

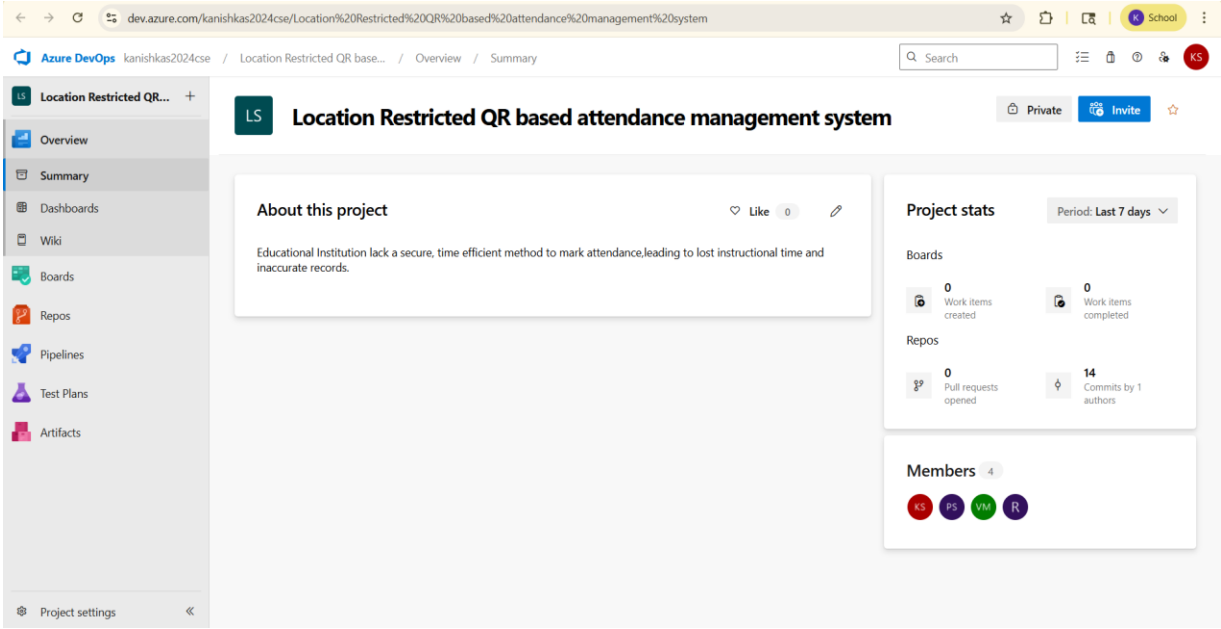
Step 6: Enable Continuous Integration

- Every time code is pushed to GitHub:
  - Pipeline triggers automatically
  - Application updates in Azure

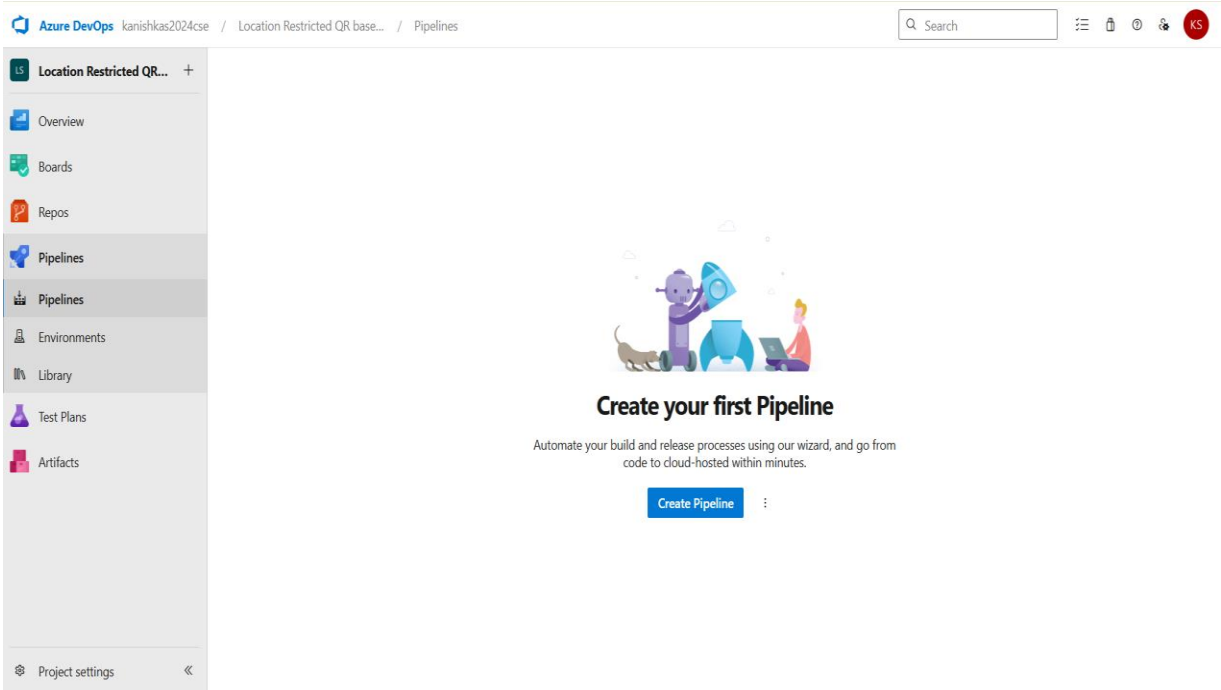
RESULT

The CI/CD pipeline was successfully created in Azure DevOps. The application was automatically built and deployed whenever changes were pushed to the repository.

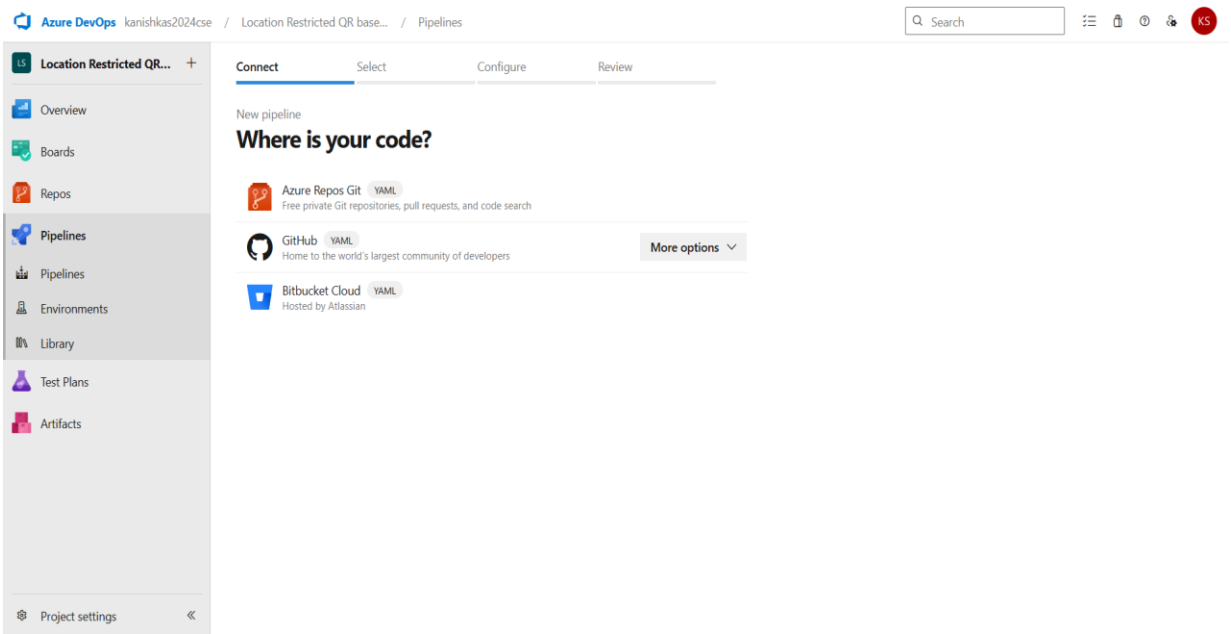
Attendance Management System in Azure



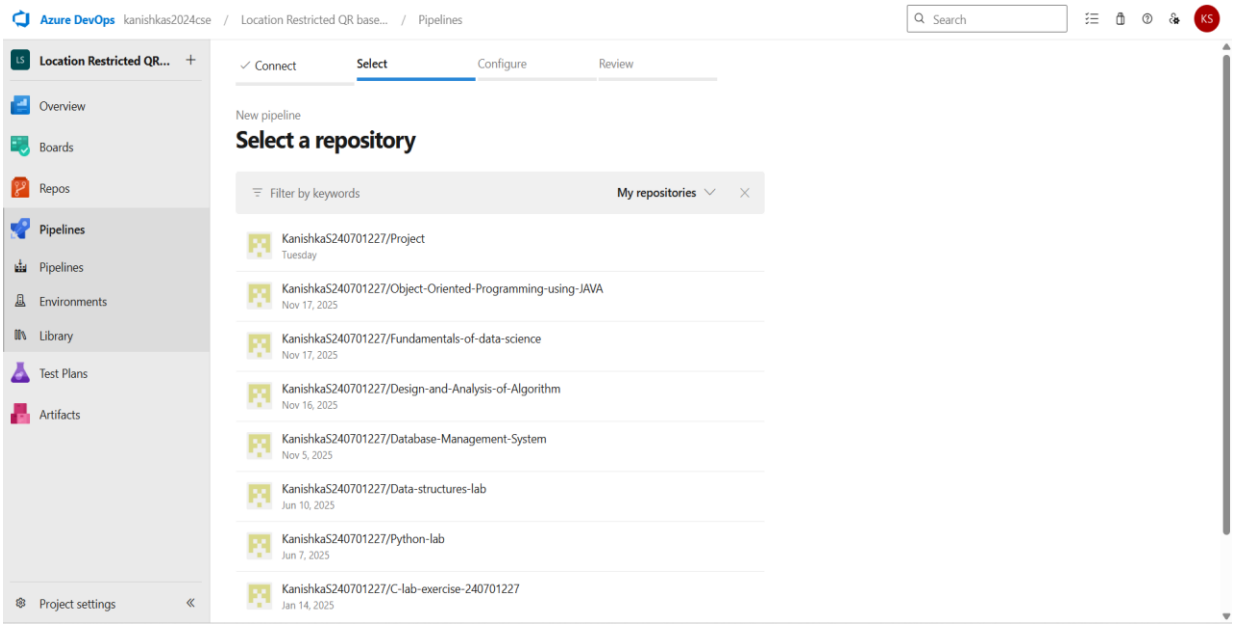
Select Pipeline



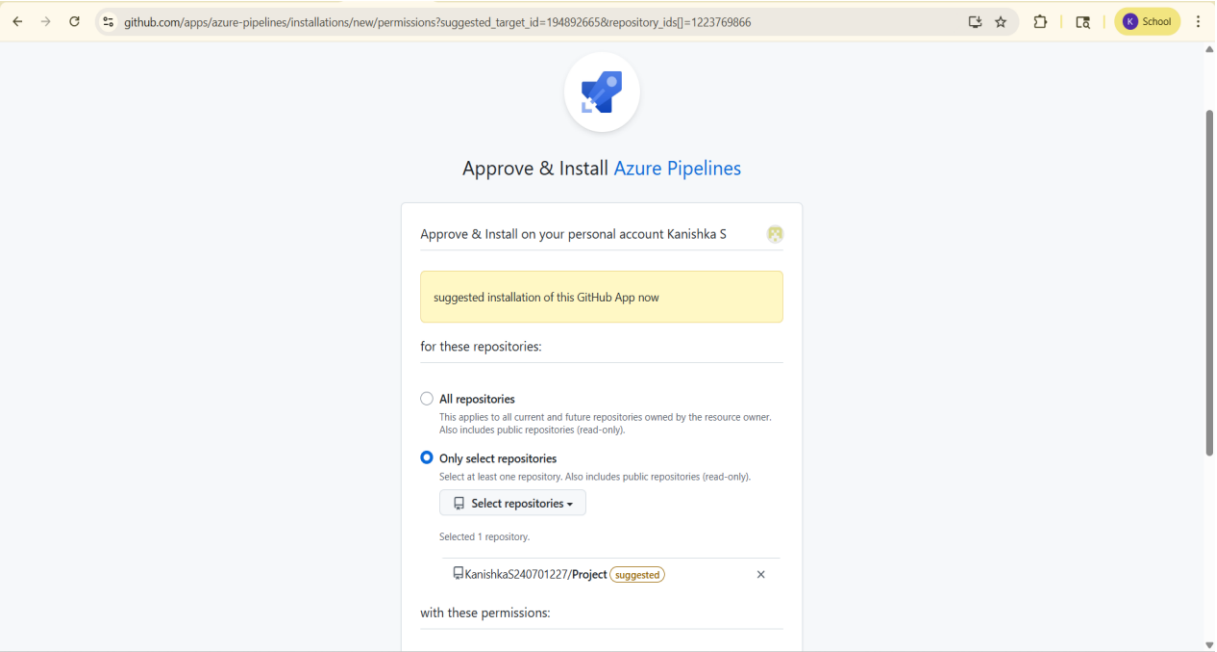
After Clicking Create Pipeline

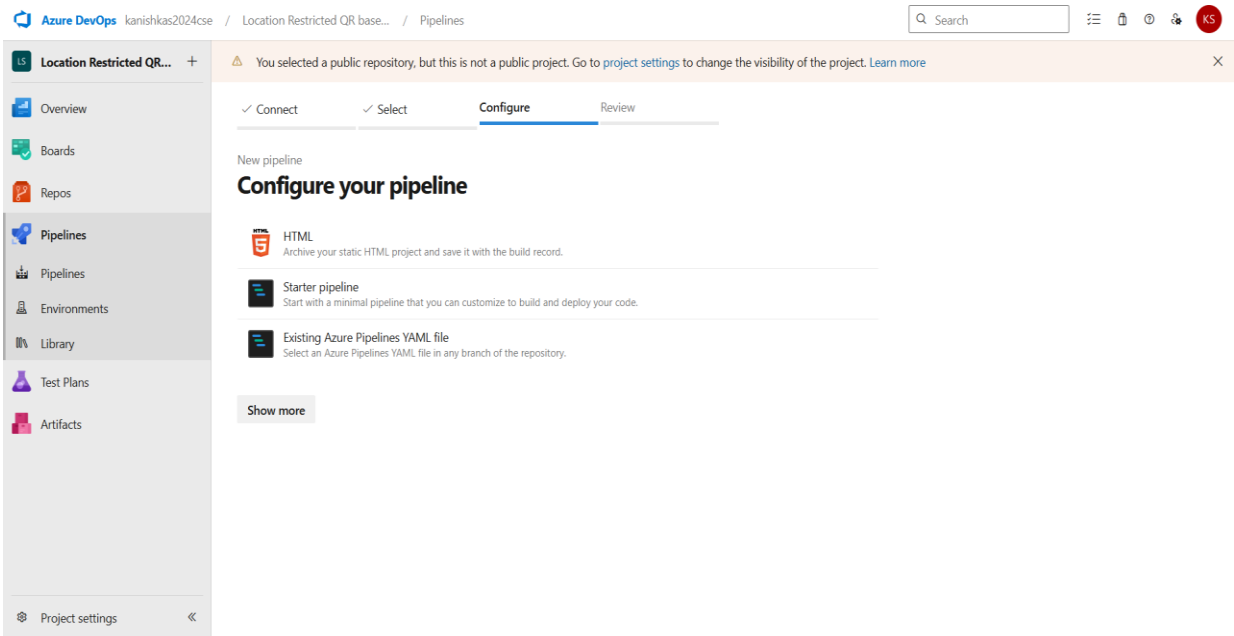


1. Select GitHub
2. After choosing GitHub
3. Select a Repository appears



Choose Online Quiz Management Repository





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Project settings

You selected a public repository, but this is not a public project. Go to project settings to change the visibility of the project. Learn more

✓ Connect ✓ Select ✓ Configure Review

New pipeline

Review your pipeline YAML

Variables Save and run

KanishkaS240701227/Project / azure-pipelines.yml *

Show assistant

```yaml
# Starter pipeline
# Start with a minimal pipeline that you can customize to build and deploy your code.
# Add steps that build, run tests, deploy, and more:
# https://aka.ms/yaml

trigger:
- main

pool:
  vmImage: ubuntu-latest

steps:
- script: echo Hello, world!
  displayName: 'Run a one-line script'

- script: |
    echo Add other tasks to build, test, and deploy your project.
    echo See https://aka.ms/yaml
  displayName: 'Run a multi-line script'
```

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Save and run

Saving will commit azure-pipelines.yml to the repository.

Commit message

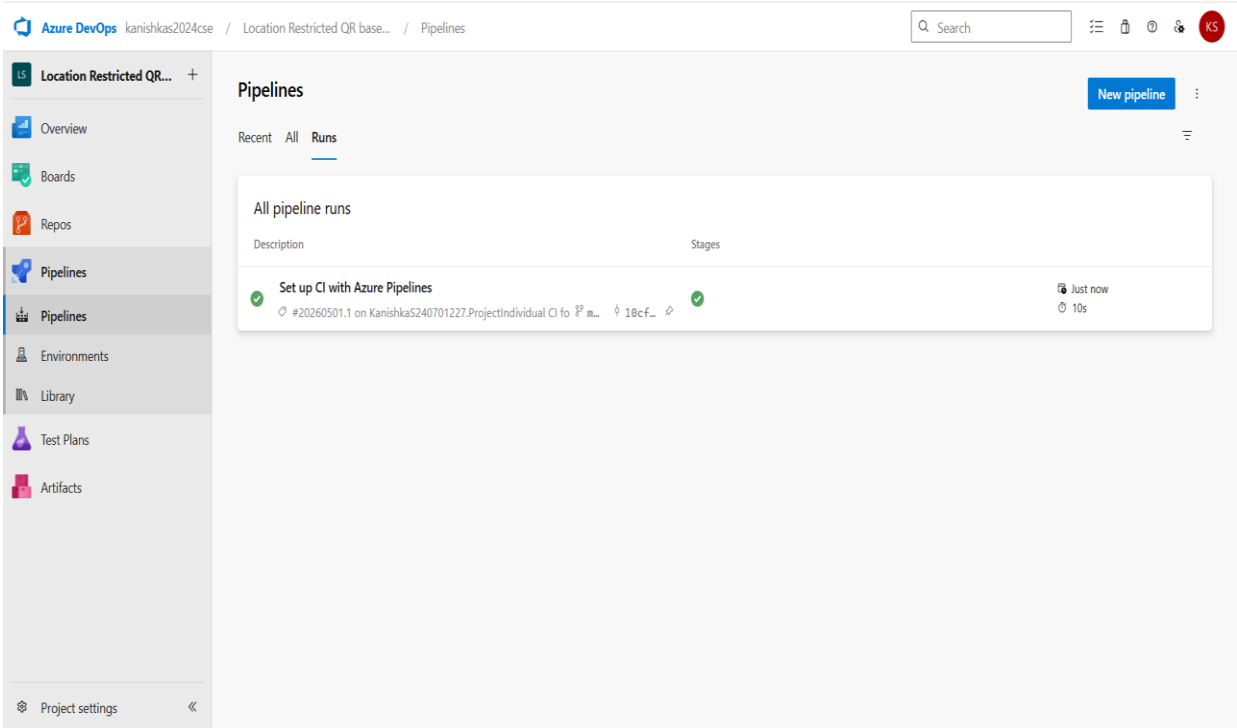
Set up CI with Azure Pipelines

Optional extended description

Add an optional description...

- (●) Commit directly to the main branch
- ( ) Create a new branch for this commit

Save and run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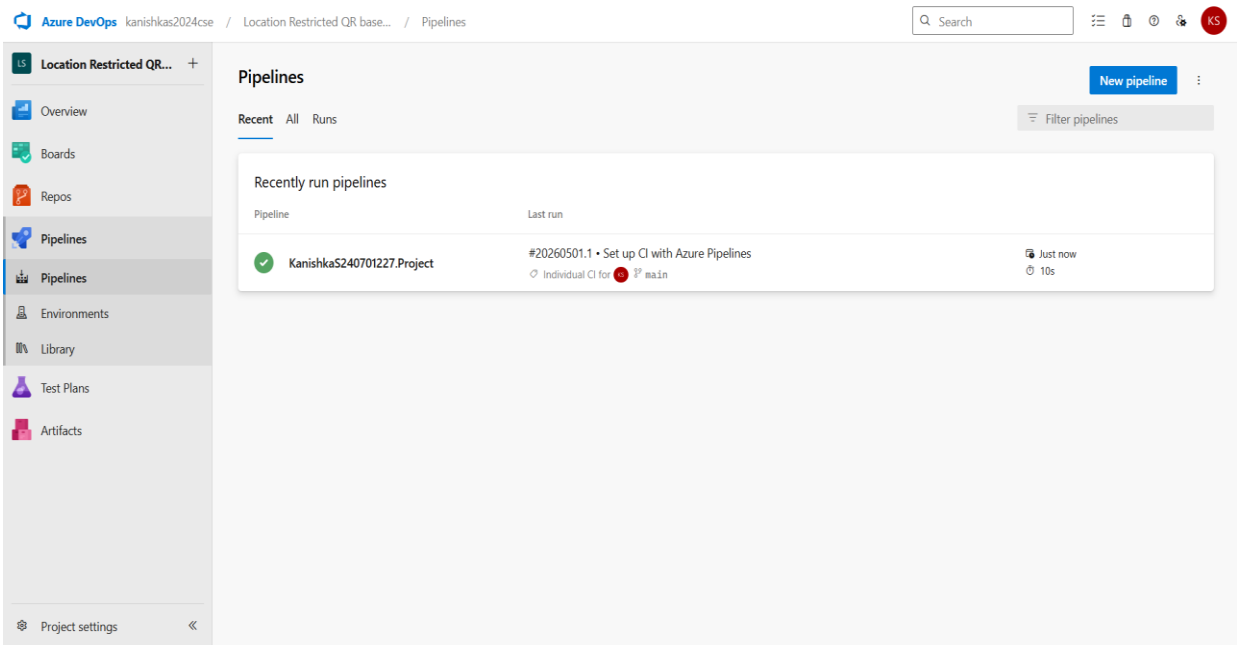
Recent All Runs
All pipeline runs
Description
Stages
Set up CI with Azure Pipelines
#20260501.1 on KanishkaS240701227.ProjectIndividual CI fo m... 10cf...
Just now
10s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#20260501.1 • Set up CI with Azure Pipelines
Individual CI for KS main
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10s